

The Pipe Fitter App — finding the common measurements

Every screen is reached from the home list. Type what you measured; read what you cut.

1. Offset around an obstruction — 45°

The one you do most.

Simple offset → **Offset** = how far you have to move over → tap preset **45°**.

Read **PIPE CUT**. That is the piece between the two fittings, gap already off.

12" offset, 2" pipe, 45° → travel 16.97, setback 1.24 each end, **cut 14.30** (14.49 before the two 3/32" weld gaps come off)

Run fills itself in: at 45° it equals the offset.

2. Offset at any other angle

Same screen. Presets are **11.25° · 22.5° · 30° · 45° · 60° · 90°**, or type any angle into **Fitting angle**.

3. Square jog — 90°

Preset **90°**. The pipe goes straight across and advances nothing.

Run = 0. Travel = the offset.

24" offset, 2" pipe → **cut 17.81** (24 – 3 – 3 – two 3/32" gaps)

4. When the run is fixed, not the angle

You know how much room you have along the line, and the angle has to fall out of it.

Simple offset → **Offset** → tap **Set run** → type the run.

The angle is solved for you and shown as **Cut angle**. A run of **0** gives you the square jog.

5. Rolling offset — moves two ways at once

Rolling offset → **Rise / set** (up or down) → **Roll** (left or right) → tap an **Elbow angle** preset.

- **True offset** — the real diagonal distance, straight through
- **Cut angle** — what you set the fittings to
- **PIPE CUT** — the piece between them

12" up, 9" over → true offset **15"**, roll angle **36.87°**

To fix the run instead of the angle, tap **Run** and type it.

6. Cut length between two fittings

You have a centre-to-centre dimension off a drawing and a fitting on each end.

Cut length → **C2C length** → pick **Joint** (screwed / welded / flanged / soldered) → pick **End A** and **End B**.

Read **PIPE CUT**.

24" C2C, 2" pipe, two long radius 90s → cut **18"**

Flanged adds a **Class** row — 150, 300, 600 and up. Anything not on the list: choose **Custom** and type the takeout.

7. Screwed pipe — how far it makes up

Thread engagement → **Size** → **C2C length**.

Gives makeup, takeout and tap drill. Override the fitting with **Fitting C-to-face** if yours is not standard.

8. Bending your own pipe

Pipe bend → radius and angle → setback, arc length and gain, with the bend marks.

Saddle bend for three and four point saddles. **Miter bend** for a segmented elbow — it checks the B31.3 22.5° limit for you.

9. Looking a figure up

Handbook → search. Every table carries its page number.

10. Checking the app in your head

Multiply the offset by these. The app agrees to four decimal places — it is tested against them.

Fitting angle	Travel	Run	Shrink
11.25°	× 5.126	× 5.027	× 0.098
22.5°	× 2.613	× 2.414	× 0.199
30°	× 2.000	× 1.732	× 0.268
45°	× 1.414	× 1.000	× 0.414
60°	× 1.155	× 0.577	× 0.577
90°	× 1.000	× 0	× 1.000

Travel = centre to centre of the two fittings. **Run** = how far the offset advances along the line. **Shrink** = travel – run, what the offset costs you.

11. The one trap — a bend is not a bought fitting

They are the same at 90° and different everywhere else.

2" long radius 45	Takes out
Bent pipe	1.2426"
Bought elbow	1-3/8"

A run of bought fittings worked to the bend formula cuts **every piece an eighth long**.

The **Cut length** screen offers both, labelled. The **Simple offset**, **Rolling offset** and **Pipe bend** screens work the bend, which is what a bender does.

Putting it on someone else's phone

Send them the web link. Works on iPhone and Android, no store, nothing to sign.

iPhone — open the link in **Safari** (it has to be Safari) → **Share** → **Add to Home Screen**.

Android — open in Chrome → ⋮ → **Add to Home screen** / **Install app**.

It then opens from an icon like any other app, and works with no signal. It updates itself the next time they open it after you push a change.

The calculator reads in inches

It stays in inches. 30 inches is 30" , not 2' 6" . The **IN** marker top-left says so.

- **Feet** converts what is on display to feet and inches — 2' 6" , not a decimal. The marker changes to **FT-IN**.
 - **Inch** takes it back.
 - The choice sticks until you change it, and survives a Clear.
 - **Settings** → **Length readout** sets which one it opens on.
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Settings worth setting once

- **Weld gap** — default 3/32". Comes off welded and flanged ends only; a screwed or soldered joint pulls up tight and never has it deducted.
 - **Length readout** — Inches (default) or Feet + inches, for the calculator.
 - **Fractions** — off, 1/8, 1/16, 1/32, 1/64. The decimal under it is always the exact figure. Cut to the decimal when tolerance is tight.
 - **Default size** — the pipe every screen opens on.
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Bolting a flange up

Flange bolt-up → set the class and size, and the bolt count follows.

- Three passes **across** the flange at about **30%**, **60%** and **100%**, then a fourth pass **round** it at full torque. The fourth is the one that gets skipped and the one that matters.
 - Every bolt in a cross pass is followed by the bolt **straight across** from it.
 - 8 bolts: 1 5 3 7 2 6 4 8 . 16 bolts: 1 9 5 13 3 11 7 15 2 10 6 14 4 12 8 16 .
 - Tap each bolt as you tighten it. **Grey** → **yellow** → **orange** → **blue** → **green**, one step per pass.
 - Tap the wrong bolt and nothing happens: it flashes red on the **right** one and stays put.
 - Put the job's final torque in the box and it shows the ft-lb for each pass. It will not guess the figure for you.
 - Bolt 1 is the first hole clockwise of top dead centre, half a pitch off the centreline — which is how flanges are drilled.
 - **Joint complete** means the sequence was followed, not that the torque was measured. Re-check anything that runs hot once it is up to temperature.
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The joint register

Joint register on the home screen, or the list icon in the flange bolt-up bar.

- Every bolt-up is written down **on every bolt**. Close the app mid-pass and you come back on the same bolt.
 - The **unnamed joint** is saved too. You never have to name anything.
 - **Name it** keeps a joint by tag, with the work already done, and clears the unnamed slot for the next one.
 - Four bars on a row are the four passes, in the same yellow/orange/blue/green as the face.
 - **Part done** at the top, **Finished** below. Delete confirms in the row.
 - Changing the size, class or bolt count **starts that joint again** — a sixteen-bolt pattern means nothing on a twelve-bolt flange.
 - It lives on that phone. It does not sync, and two people on one flange keep two lists.
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The re-torque log

On a finished joint, under **Re-torque**.

- A joint that was right **cold can be slack hot**: the gasket creeps, the bolts and flanges grow at different rates, the whole thing relaxes.
 - Once the line has been up to temperature and back, go round at full torque and tap **Record a check**.
 - The question is **Bolts moved** or **All tight**, with no default. That finding is the reading, not the date.
 - **Bolts took up** = still relaxing, go back after another cycle. **All tight** = settled, and that closes it out.
 - The register splits into **Needs a re-check** and **Closed out**. Work the first list after a startup.
 - **Clear closed out** never touches a joint still waiting on a check.
 - The app will not tell you *when* to go back — that comes from the job's spec, not from an app.
 - Reopening a bolt-up, or changing the flange, throws its log away with it.
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The 3D spool

3D spool → add legs. A leg is **a length and a direction** — 36" east, 24" straight up, 30" north. No bend angle, no roll.

- The app works out every turn and names the fitting: **Turns 90° off leg 1 – 90° elbow**. A turn that is not stock is flagged above the leg list.
- Every leg on the picture carries its length and direction; every fitting carries its angle.
- **Nine views**: the four isometric corners, **Plan**, and an elevation from each side.
- It **opens on the corner this spool reads best from** — the one with no legs crossing and nothing lost end-on.
- **Drag** to turn it: pull right and it goes right, pull down and the top comes over. The yaw goes the whole way round.
- The **tilt stays between 15° and 75°** above level — the band where all three directions still read. Under the spool and straight-down are not hand-turn views; the **Plan** and **elev** buttons give you those deliberately.
- A leg drawn as a **circle** is coming straight at you. Read its figure — that is what a plan is.

- **Hold a leg** in the picture and drag along it to stretch or shorten it. Pull the way it runs for longer.
- While you pull, the drawing **holds its scale and its place** — the leg follows your thumb, its neighbours stay put. Letting go refits the whole run.
- **Mirror** gives the opposite hand; **Turn over** swaps every rise for a drop; **Swing 90°** points the whole spool elsewhere. None of the three changes a cut.
- All three sit **under the drawing**, and the spool **swings into place over half a second** so you can watch it. The line below names what just happened.

Saved spools — under the handing buttons.

- **Save this spool** keeps the legs, pipe and gap on the phone, by name. Survives restarts and updates.
- Tap one to open it. The header says *saved* or *unsaved changes*.
- Same name updates it; a **new name keeps both**. Delete takes two taps.
- A shelf written by a newer app version is never overwritten by an older one.

Share drawing — under the picture, next to Save.

- One page: the spool in three dimensioned views, the cut list, the elbows, and the sticks to pull.
- Phone → PDF to the share sheet. Browser → the print dialog.
- A line drawing, not the shaded screen picture: hairlines, black on white, weld ticks, open ends capped.
- The isometric is the corner it reads best from; a leg drawn as a circle is coming at you — read its figure.
- Unsaved spools print as *Spool*. Save it first to get its name on the sheet.

Cut list — under the elbows. How many sticks to pull, what comes off each, and what is left.

- Packed onto **Settings** → **Stock length**, with **Settings** → **Saw cut** charged on every piece.
- It finds the **fewest** sticks, and among those the one that leaves the **longest single drop**.
- A cut longer than a stick is named, not quietly planned around.
- Cut length, simple offset and rolling offset now say whether their one cut comes off one stick, and what is left.

The order sheet

Order sheet → one order across every saved spool, instead of one order per spool.

- Every spool ordered alone rounds up to a whole stick. Ordered together they round up **once**. On six spools that is typically **one stick in five**.
- It opens with **every saved spool ticked**. Tap one off and every figure moves — that is how to see what a spool is worth to the order.
- Three figures up top: **what to buy**, what **one at a time** would cost, and the **difference**.
- Each spool takes a **letter**, and every piece is stamped letter-and-leg: **A3** is leg three of spool A. The letters are on the pick list, on the sticks, and in the key on the sheet.
- **Pipe does not pool across sizes or schedules** — 2" SCH 40 and 2" SCH 80 are different sticks, packed and listed separately, biggest pipe first.

- **Radius does pool.** LR and SR change the cuts, not the pipe the cuts come off. The key says which is which.
- **Share order sheet** — top half for whoever buys the pipe (size, schedule, sticks, total length), bottom half for the saw.
- A spool that will not build is **listed with its reason**, never silently left off. A piece longer than a stick names its mark, so you know which spool to fix.
- If everything already fills its own sticks it says so. Nothing wasted means nothing to recover.